

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	1 October 2026
Team ID	SWTID-2026-8511
Project Name	Implement Client Script and UI Policy (Incident)
Maximum Marks	4 Marks

Technical Architecture:

Architecture of the Incident form with Client Script and UI Policy on ServiceNow, followed by Table-1 and Table-2.

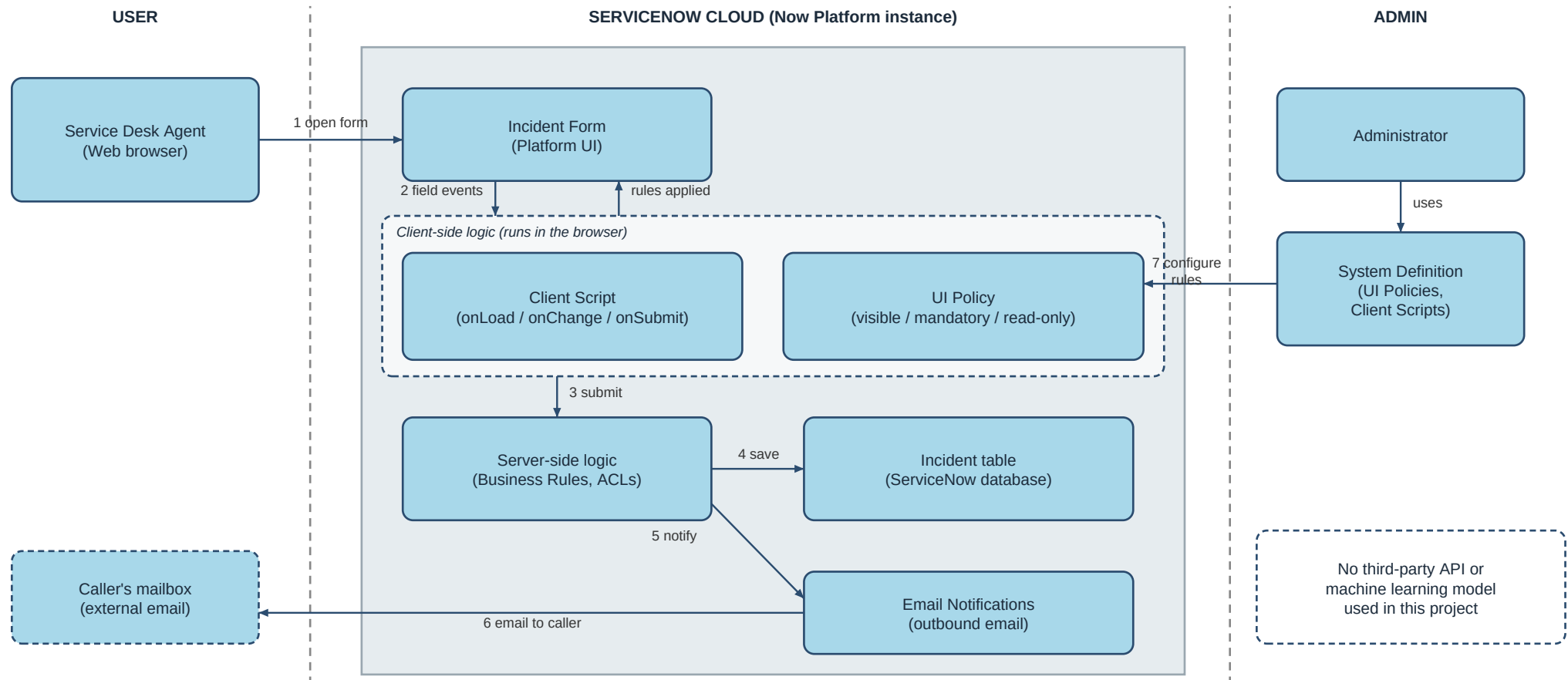


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	How the service desk agent interacts with the application	ServiceNow Platform UI (Incident form in the browser), HTML, CSS, JavaScript
2	Application Logic-1	Client Script: guides and validates input on the form (onLoad, onChange, onSubmit)	JavaScript (ServiceNow Client Script, g_form API)
3	Application Logic-2	UI Policy: shows or hides fields and makes them mandatory or read-only	ServiceNow UI Policy (conditions and UI Policy Actions)
4	Application Logic-3	Server-side logic that still applies rules when data is saved	ServiceNow Business Rules and ACLs (server-side JavaScript)
5	Database	Stores the incident records and their field values	ServiceNow Incident table (incident)
6	Cloud Database	Database service on the cloud	ServiceNow-hosted instance database (managed by the platform)
7	File Storage	Attachments added to an incident (if any)	ServiceNow attachment storage
8	External API-1	Email notification sent to the caller about the incident	ServiceNow outbound email (SMTP)
9	External API-2	Not used in this project	N/A
10	Machine Learning Model	Not used in this project	N/A
11	Infrastructure (Server / Cloud)	Application deployment on a cloud instance. Local server configuration: not required. Cloud server configuration: ServiceNow instance	ServiceNow Personal Developer Instance (PDI), cloud-hosted

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open-source technologies used with the platform	Standard JavaScript, HTML and CSS; ServiceNow client APIs (g_form, g_user)
2	Security Implementations	Access controls that protect the rules and the data	Role-based access (administrator role for scripts), ACLs, HTTPS/TLS, server-side validation as backup
3	Scalable Architecture	Multi-tier platform (browser, application, database). Rules can be reused for other tables and fields	ServiceNow multi-tier cloud architecture
4	Availability	The rules run whenever the cloud instance is available. A rule can be deactivated quickly if needed	ServiceNow-managed cloud instance, Active flag on rules
5	Performance	Rules run in the browser without a server round trip. UI Policy is preferred over scripts and Client Scripts stay lightweight	UI Policy first, lightweight Client Scripts

References:

<https://c4model.com/>
<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>
<https://www.ibm.com/cloud/architecture>
<https://aws.amazon.com/architecture>
<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>